RACE
Regular Analysis Through Continuous Exercise

Three Dimensional Coordinate Geometry : RACE-01

Single Correct Question

1. The radius of sphere which passes through the point (0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1) is -

(A) $\frac{1}{2}$ (B) 1 (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}}{2}$

MTD225

Subjective Type Question

2. Find the image of the given point in the specified plane.
(i) (–3,4,7) in the YZ-plane (ii) (–7, 2,–1) in the XZ-plane

MTD226

3. Find the distance between the points $P(3, 4, 5)$ and $Q(-1, 2, -3)$.

MTD227

4. Show that the points $A(0, 7, 10)$, $B(-1, 6, 6)$ and $C(-4, 9, 6)$ are vertices of an isosceles right angled triangle.

MTD228

5. Find the locus of a point such that the difference of the squares of its distances from the points $A(3, 4, 5)$ and $B(-1, 3, -7)$ is equal to $2k^2$.

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MTD229

6. Find the co-ordinates of points which trisects the line joining the points $A(-3, 2, 4)$ and $B(0, 4, 7)$

MTD230

7. Find the ratio in which the following planes divide the line joining the points $P(-2, 4, 7)$ and $Q(3, -5, 8)$: (a) xy-plane (b) yz-plane

MTD231

8. Find the perpendicular distance of the point $P(a, b, c)$ from the coordinate axes.

MTD232

9. A point R with x co-ordinate 4 lies on the line segment joining the points $P(2,-3,4)$ and $Q(8,0,10)$. Find the coordinates of the point R.

MTD233

10. Three consecutive vertices of a parallelogram are (4,3,5), (0,6,0) and (–8,1,4). Find possible coordinates of the fourth vertex.

MTD234

11. If a line makes α, β and γ angle with axes, then prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

MTD235

12. If ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 are the direction cosines of two mutually perpendicular lines, show that the direction cosines of the line perpendicular to both of these are
$m_1n_2 - m_2n_1, n_1\ell_2 - n_2\ell_1, \ell_1m_2 - \ell_2, m_1$.

MTD236

For More Join: @IITJEE_Advanced

13. Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to (1, –2, –2) & (0, 2, 1).

MTD237

14. Find the angle between the two straight lines whose direction cosines ℓ, m, n are given by $2\ell + 2m - n = 0$ and $mn + n\ell + \ell m = 0$.

MTD238

15. Find the acute angle between two lines such that the direction cosines ℓ, m, n of each of them satisfy the equations $\ell + m + n = 0$ and $\ell^2 + m^2 - n^2 = 0$.

MTD239

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ANSWER KEY

1.	D	**2.**	(i) (3, 4, 7); (ii) (–7, –2, –1)		
3.	$2\sqrt{21}$	**5.**	$(8x + 2y + 24z + 9 + \lvert 2k^2 \rvert = 0)$		
6.	$\left(-2, \frac{8}{3}, 5\right)$ and $\left(-1, \frac{10}{3}, 6\right)$	**7.**	(a) 7 : 8 externally; (b) 2 : 3 internally		
8.	$\sqrt{b^2+c^2}$; $\sqrt{a^2+c^2}$ and $\sqrt{a^2+b^2}$	**9.**	4, –2, 6	**10.**	–4, –2, 9 or 12, 8, 1
13.	$\left(\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}\right)$ or $\left(\frac{-2}{3}, \frac{1}{3}, \frac{-2}{3}\right)$	**14.**	90°	**15.**	60°

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Three Dimensional Coordinate Geometry : RACE-02

Single Correct Question

1. The line $\frac{x-x_1}{0}=\frac{y-y_1}{1}=\frac{z-z_1}{2}$ is

(A) parallel to x-axis

(B) perpendicular to x-axis

(C) perpendicular to YOZ plane

(D) parallel to y-axis

MTD240

Multiple Correct Question

2. Which of the following points lie on the line $2x - 2 = 3y + 6 = 4 - 4z$.

(A) (1,–2,1) (B) (7,2,–2) (C) (–11,–10,7) (D) (–5,–6,4)

MTD241

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Subjective Type Question

3. Find the equation of the line passing through the point (4, 2, 3) and having direction ratios 1,–1, 2.

MTD242

4. Find the vector form of line whose cartesian equation is $\frac{x-5}{3}=\frac{y+4}{7}=\frac{z-6}{2}$.

MTD243

5. Find the equation of line passing through (–1,0,2) and (3,4,6) in both cartesian and vector form.

MTD244

6. Find the equation of the line through (3,1,2) and equally inclined to the axes.

MTD245

7. Find direction cosines of line $2x + 1 = 1 - 6y = 4z + 5$.

MTD246

8. Find equation of line passing through (–1,2,3) and parallel to line $x = -2y = 3z$.

MTD247

9. Find the equation of the line passing through (3,–1,11) and perpendicular to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$

MTD248

For More Join: @IITJEE_Advanced

10. If the foot of the perpendicular drawn from the point (1,0,3) on a line passing through (α,7,1) is $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$, then α is equal to

MTD249

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ANSWER KEY

1.	B	**2.**	A,B,C,D
4.	$\vec{r}=\left(5\hat{i}-4\hat{j}+6\hat{k}\right)+\lambda\left(3\hat{i}+7\hat{j}+2\hat{k}\right)$	**5.**	$\vec{r}=\left(-\hat{i}+2\hat{k}\right)+\lambda\left(4\hat{i}+4\hat{j}+4\hat{k}\right)$ and $\frac{x+1}{1}=\frac{y}{1}=\frac{z-2}{1}$
6.	$\frac{x-3}{\pm\frac{1}{\sqrt{3}}}=\frac{y-1}{\pm\frac{1}{\sqrt{3}}}=\frac{z-2}{\pm\frac{1}{\sqrt{3}}}$	**7.**	$\ell=\pm\frac{6}{7}$; $m=\mp\frac{2}{7}$; $n=\pm\frac{3}{4}$
8.	$\frac{x+1}{0}=\frac{y-2}{-\frac{1}{2}}-\frac{z-3}{\frac{1}{3}}$	**9.**	$\frac{x-3}{1}=\frac{y+1}{-6}=\frac{z-11}{4}$
10.	4		

For More Join: @IITJEE_Advanced

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Three Dimensional Coordinate Geometry : RACE-03

Single Correct Question

1. If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar, then k can have :

(A) any value (B) exactly one value

(C) exactly two values (D) exactly three values.

MTD250

2. The shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$ is

(A) $\frac{7}{2}\sqrt{30}$ (B) $3\sqrt{30}$ (C) 3 (D) $2\sqrt{30}$

MTD251

Multiple Correct Question

3. If the lines $\frac{x-1}{2}=\frac{y+1}{k}=\frac{z}{2}$ and $\frac{x+1}{5}=\frac{y+1}{2}=\frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is(are)

(A) $y + 2z = -1$ (B) $y + z = -1$ (C) $y - z = -1$ (D) $y - 2z = -1$

MTD252

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Subjective Type Question

4. Find p, if lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are perpendicular.

MTD253

5. Find angle between lines $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z+3}{2}$ and $\frac{x}{1}=\frac{y}{1}=\frac{z}{0}$

MTD254

6. Prove that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ are coplanar. Find their point of intersection.

MTD255

7. A line with d.r.'s 2,1,2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$ at A and B respectively. Then find the coordiantes of A and B.

MTD256

8. Find the shortest distance between the lines $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5};\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$.

MTD257

For More Join: @IITJEE_Advanced

9. Find the shortest distance between the lines $\vec{r}=\left(2\vec{i}-\vec{j}-\vec{k}\right)+\lambda\left(2\vec{i}+\vec{j}+2\vec{k}\right)$ and $\vec{r}=\left(\vec{i}+2\vec{j}+\vec{k}\right)+\mu\left(\vec{i}-\vec{j}+\vec{k}\right)$.

MTD258

10. Find the shortest distance between lines $\vec{r}=2\vec{i}+3\vec{j}-\vec{k}+\lambda\left(\vec{i}-\vec{j}+2\vec{k}\right)$ and $\vec{r}=-3\vec{i}+4\vec{j}+\vec{k}+\mu\left(\vec{i}-\vec{j}+2\vec{k}\right)$.

MTD259

11. A line with d.r's (2,7,–5) is drawn to intersect the lines $\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}$ and $\frac{x+3}{-3}=\frac{y-3}{2}=\frac{z-6}{4}$ at P and Q respectively. Find PQ

MTD260

12. Find the equation of line of shortest distance between the lines $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$; $\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$

MTD261

13. If the two lines $\ell_1=\frac{x-2}{3}=\frac{y+1}{-2}$, $z=2$ and $\ell_2:\frac{x-1}{1}=\frac{2y+3}{\alpha}=\frac{z+5}{2}$ perpendicular, then find the angle between the lines ℓ_2 and $\ell_3:\frac{1-x}{3}=\frac{2y-1}{-4}=\frac{z}{4}$.

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MTD262

ANSWER KEY

1.	C	**2.**	B	**3.**	B,C	**4.**	$p=\frac{70}{11}$	**5.**	45° or 135°
7.	$A(3a, 2a, 3a)$ and $B(a, a, a)$			**8.**	0			**9.**	$\frac{3}{\sqrt{2}}$
10.	$\frac{2\sqrt{22}}{\sqrt{3}}$	**11.**	$\sqrt{78}$	**12.**	$\frac{x-3}{4}=\frac{y-5}{6}=\frac{z-7}{8}$			**13.**	$\theta=\sec^{-1}\frac{29}{4}$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Three Dimensional Coordinate Geometry : RACE-04

Single Correct Question

1. Consider three vectors $\vec{p}=\hat{i}+\hat{j}+\hat{k}, \vec{q}=2\hat{i}+4\hat{j}-\hat{k}$ and $\vec{r}=\hat{i}+\hat{j}+3\hat{k}$. If $\vec{p}$, $\vec{q}$ and $\vec{r}$ denotes the position vector of three non-collinear points then the equation of the plane containing these points is

(A) $2x - 3y + 1 = 0$ (B) $x - 3y + 2z = 0$

(C) $3x - y + z - 3 = 0$ (D) $3x - y - 2 = 0$

MTD263

Subjective Type Question

2. Equation of a plane is $3x + 4y + 5z = 7$. Find

(a) The direction cosines of its normal

(b) The points where it intersects the axes.

(c) Its intercept form.

(d) Its equation in normal form (in cartesian as well as in vector form)

MTD264

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3. A variable plane passes through a fixed point (1, 2, 3). Show that the locus of the foot of the perpendicular drawn from origin to this plane is given by the equation $x^2 + y^2 + z^2 - x - 2y - 3z = 0$.

MTD265

4. Find the equation of the plane through the points (2, 1, –1) and (–1,3,4) and perpendicular to the plane $x - 2y + 4z = 1$.

MTD266

5. Find the equation of the plane passing through the points (2, 1, 0), (5, 0, 1) and (4, 1, 1).

MTD267

6. Find the equation of plane passing through (1,1,1) and parallel to the lines L_1, L_2 having direction ratios (1,0, 1), (1, 1,0). Find the volume of tetrahedron formed by origin and the points where these planes intersect the coordinate axes.

MTD268

7. Find equation of plane passing through (0,0,0), parallel to $x = 2y = 3z$ and $2x = 3y = 4z$.

MTD269

8. Find equation of plane containing $x = 2y - 2 = 3z + 6$ and perpendicular to $x + 2y + 3z = 6$.

MTD270

9. Find the equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y + z = 3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, –1).

MTD271

For More Join: @IITJEE_Advanced

10. Prove that if a plane has the intercepts a, b, c and is at a distance of p units from the origin, then $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{p^2}$.

MTD272

11. Find the image of the point with position vector $3\hat{i}+\hat{j}+2\hat{k}$ in the plane $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=4$

MTD273

12. Find the image of point $P(1, 3, 2)$ in the plane $2x - y + z + 3 = 0$ as well as the foot of the perpendicular drawn from the point (1, 3, 2).

MTD274

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ANSWER KEY

1.	D	**4.**	$18x + 17y + 4z = 49$	**5.**	$x + y - 2z = 3$	**6.**	$-x + y + z = 1$ and $\frac{1}{6}$
7.	$x - 6y + 6z = 0$	**8.**	$5x - 16y + 16 + 9z + 18 = 0$	**9.**	$-9x + 21y - z + 31 = 0$		
11.	1, 2, 1	**12.**	$\left(-\frac{1}{3},\frac{11}{3},\frac{4}{3}\right)$				

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Three Dimensional Coordinate Geometry : RACE-05

Single Correct Question

1. The plane through the intersection of the planes $x + y + z = 1$ and $2x + 3y - z + 4 = 0$ and parallel to y-axis also passes through the point :

(A) $(-3,0,-1)$ (B) $(3,3,-1)$ (C) $(3,2,1)$ (D) $(-3,1,1)$

MTD275

2. A plane passing through the points $(0,-1,0)$ and $(0,0,1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$, also passes through the point

(A) $\left(-\sqrt{2},1,-4\right)$ (B) $\left(\sqrt{2},1,4\right)$ (C) $\left(\sqrt{2},-1,4\right)$ (D) $\left(-\sqrt{2},-1,-4\right)$

MTD276

Subjective Type Question

Rankers Academy JEE

3. Prove that the planes $3x - 2y + z + 17 = 0$ and $4x + 3y - 6z - 25 = 0$ are perpendicular.

MTD277

4. Find the angle between the planes $3x + 4y + z + 7 = 0$ and $-x + y - 2z = 5$

MTD278

5. Find the distance between the planes $2x - y + 3z - 4 = 0$and $6x - 3y + 9z + 13 = 0$.

MTD279

6. Find the angle between the planes $\vec{r}.\left(2\hat{i}-3\hat{j}+4\hat{k}\right)=1$ and $\vec{r}.\left(-\hat{i}+\hat{j}\right)=4$.

MTD280

7. Find the equation of the plane containing the line of intersection of the planes $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through the point (1, 1, 1).

MTD281

8. Find the equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to x-axis.

MTD282

9. Find the symmetrical form of the line $x - y + 2z = 5$, $3x + y + z = 6$.

MTD283

For More Join: @IITJEE_Advanced

10. A tetrahedron has vertices $P(1, 2, 1)$, $Q(2, 1, 3)$, $R(-1,1,2)$ and $O(0, 0, 0)$. Find the angle between the faces OPQ and PQR.

MTD284

11. Let L be the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$. If L makes an angle α with the positive X-axis then find $\cos\alpha$.

MTD285

12. Find volume of tetrahedron formed by planes $x + y + z = 3$, $x + 2y + 3z = 6$, $x + 2y + z = 4$ and $x = 0$.

MTD286

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ANSWER KEY

1.	C	**2.**	B	**4.**	$\cos^{-1}\frac{1}{\sqrt{156}}$	**5.**	$\frac{25}{3\sqrt{14}}$		
6.	$\frac{5}{\sqrt{58}}$	**7.**	$20x+23y+26z-69=0$			**8.**	$-y+3z-6=0$		
9.	$\frac{x-\frac{11}{4}}{-3}=\frac{y-\frac{11}{4}}{5}=\frac{z-0}{4}$	**10.**	$\left(\cos^{-1}\frac{19}{35}\right)$	**11.**	$\frac{1}{\sqrt{3}}$				
12.	$\frac{1}{6}$								

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Three Dimensional Coordinate Geometry : RACE-06

Single Correct Question

1. The line $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is perpendicular to which of the following planes?

(A) $2x + y - 2z = 0$ (B) $3x + 4y + 5z = 7$

(C) $x + y + z = 2$ (D) $2x + 3y + 4z = 0$

MTD287

Subjective Type Question

2. Find the angle between the plane $3x + 4y + 5 = 0$ and the line $\frac{x-1}{2}=\frac{y-2}{0}=\frac{z-1}{1}$.

MTD288

3. If the line $\frac{x+1}{-2}=\frac{y+\lambda}{3}=\frac{z+5}{\mu}$ lies in the plane $x + 2y - z = 0$, then find λ and μ.

MTD289

4. Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$. Then find α and β.

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MTD290

5. If the angle between the line $x=\frac{y-1}{2}=\frac{z-3}{\lambda}$ and the plane $x + 2y + 3z = 4$ is $\cos^{-1}\left(\sqrt{5/14}\right)$ then find λ.

MTD291

6. Find the absolute value of m for which the line $3x - 2y + z + 3 = 4x - 3y + 4z + 1 = 0$ is parallel to the plane $2x - y + mz - 2 = 0$.

MTD292

7. If θ is the angle between line $\frac{x+1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$ and the plane $2x-y+\sqrt{\gamma}z+4=0$ is such that $\sin\theta=\frac{1}{3}$, then find γ.

MTD293

8. Find equation of a plane which passes through the point (3,2,0) and the line $\frac{x-4}{1}=\frac{y-7}{5}=\frac{z+4}{4}$.

MTD294

9. Find the equation of the plane containing the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ and perpendicular to the plane $x + 2y + z = 12$.

MTD295

For More Join: @IITJEE_Advanced

10. The point P is the intersection of the straight line joining the points $Q(2,3,5)$ and $R(1,-1,4)$ with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point $T(2,1,4)$ to QR, then find the length of the line segment PS.

MTD296

11. Find the equation of the plane containing the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$.

MTD297

12. Find the equation to the line of greatest slope through the point (7, 2, –1) in the plane $x - 2y + 3z = 0$ assuming that the axes are so placed that the plane $2x + 3y - 4z = 0$ is horizontal.

MTD298

13. **(a)** Find the projection of line $\frac{x-3}{2}=\frac{y-4}{3}=\frac{z-5}{4}$ in the plane $4x + 4y - 5z + 54 = 0$.

MTD299

(b) The projection of the line $\frac{x+1}{-1}=\frac{y}{2}=\frac{z-1}{3}$ in the plane $x - 2y + z = 6$ is the line of intersection of this plane with which of the following planes?

(A) $2x + y + 2 = 0$ (B) $3x + y - z = 2$

(C) $2x - 3y + 8z = 3$ (D) $x + y - z = 1$

MTD300

14. Find the image of the line $\frac{x-1}{3}=\frac{y-3}{1}=\frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$.

MTD301

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ANSWER KEY

1.	B	**2.**	$\frac{6}{5\sqrt{5}}$	**3.**	$\lambda = 2, \mu = 4$	**4.**	$\alpha = -6, \beta = 7$
5.	$\frac{2}{3}$	**6.**	2	**7.**	$\frac{5}{3}$	**8.**	$5x - y = 13$
9.	$-9x + 2y + 5z - 4 = 0$			**10.**	$\frac{1}{\sqrt{2}}$	**11.**	$x - 2y + z = 0$
12.	$\frac{x-7}{22}=\frac{y-2}{5}=\frac{z+1}{-4}$			**13.**	**(a)** $\frac{x+1}{2}=\frac{y-0}{3}=\frac{z-10}{4}$	**(b)**	(A)
14.	$\frac{x+3}{3}=\frac{y-5}{1}=\frac{z-2}{-5}$						

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